

**GOVINDRAM SEKSARIA INSTITUTE OF
TECHNOLOGY & SCIENCE, INDORE
END SEMESTER EXAMINATION**

B.E. — First Year (Mechanical Engineering)
Maximum Marks: 56 Time: 3 Hours

SET D

UNIT I — Mechanical Properties (14 Marks)

- Q1.** (a) Define impact strength. (2 Marks) (b) Define modulus of resilience. (2 Marks)
Q2. Explain the phenomenon of strain hardening. (3 Marks)
Q3. Explain creep curve and discuss its three stages with a neat diagram. (7 Marks)

UNIT II — Theory of Machines (14 Marks)

- Q4.** (a) Define Grashof's law. (2 Marks) (b) Define instantaneous center. (2 Marks)
Q5. Explain types of constrained motion. (3 Marks)
Q6. Explain Klein's construction for velocity analysis of slider-crank mechanism. (7 Marks)

UNIT III — Fluid Kinematics (14 Marks)

- Q7.** (a) Define vorticity. (2 Marks) (b) Define circulation. (2 Marks)
Q8. Write short note on flow net. (3 Marks)
Q9. Derive Bernoulli's equation from Euler equation. (7 Marks)

UNIT IV — IC Engines & Automobiles (14 Marks)

- Q10.** (a) Define scavenging. (2 Marks) (b) Define ignition delay. (2 Marks)
Q11. Compare SI and CI engines. (3 Marks)
Q12. Explain working of MPFI system with labelled diagram. (7 Marks)